



Effective Properties of Smart Composites using Finite Element Analysis

UG Project Report

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Keywords:

Piezoelectric composites, effective properties, micromechanics, piezoelectricity, smart materials, strength of materials.

Abstract:

Piezoelectric materials deform under an electric field and generate electric charges when mechanically stressed, making them suitable for use as both sensors and actuators. When embedded in flexible structures, they enable self-monitoring and control capabilities. Their performance can be enhanced by forming piezoelectric fiber-reinforced composites (PFRCs).

This study aims to determine the effective properties of PFRCs through micromechanical analysis. The Finite Element Method (FEM), using ANSYS, will be employed to compute these properties, which will then be compared with results from the Strength of Materials (SM) approach. The electric field is applied transversely to the fiber direction and is uniform in both fiber and matrix. Various piezoelectric fibers with different volume fractions and orientations will be tested to identify optimal composite configurations.

Introduction:

Piezoelectricity refers to the electromechanical interaction between mechanical and electrical states within certain materials, particularly piezoelectric composites. These materials can generate an electric field in response to mechanical stress or strain and, conversely, deform when subjected to an electric field (James et al., [1998](#)). They are widely used in engineering applications such as vibration and noise control, precision positioning and cutting, and microelectromechanical systems (MEMS), as well as in commercial actuators and sensors.

Research in this field primarily focuses on micromechanical modeling to estimate the mechanical and electromechanical properties of these composites. This project aims to predict the effective elastic and piezoelectric coefficients of unidirectional piezoelectric fiber-reinforced composites (PFRCs) subjected to a

constant electric field applied transversely to the fiber direction. The primary objective is to determine the effective piezoelectric coefficient responsible for inducing normal stress along the fiber direction.

Approach:

1. Design:

Firstly, a 2D symmetrical geometry of a piezoelectric composite is designed using ANSYS APDL. The composite is considered to be isotropic therefore its properties will be the same in all directions. Owing to this consideration, we have designed only a quarter part of the entire geometry and performed analysis and executed calculations on it which will remain the same for the entire material. The dimensions of the designed geometry are as follows:

Side length of the square matrix region: 10 units.

2. Material Assignment:

For matrix:

Density : 2800 kg/m^3

Stiffness coefficient = 6.59 GPa

Stiffness coefficient = 5.2 GPa

Poisson ratio= 0.3

For fiber:

Density : 7600 kg/m^3

Stiffness coefficient = 111 GPa

Stiffness coefficient = 68 GPa

$e_p^{31}, e_p^{32}, e_p^{33}$, are the piezoelectric stress coefficients

$$e_{31}^p = 3.1 \text{ C/m}^2 \quad e_{32}^p = 3.1 \text{ C/m}^2 \quad e_{33}^p = 15 \text{ C/m}^2$$

3. Meshing:

After designing the geometry, we have to create mesh over the geometry to apply the boundary conditions and perform calculations at the nodes. We have created mesh separately on the fiber and matrix geometry. The mesh size is kept at 2 units.

4. Boundary Conditions and Solving:

We have applied the following boundary conditions to our geometry to obtain the final results:

- The displacement at the bottom surface is zero in the upward direction.
- The displacement at the left surface is zero in the right direction.

We have applied boundary conditions in two steps to calculate the stiffness coefficient and the piezoelectric stress coefficient.

STEPS:

- Through this step, we calculated the stiffness coefficients.
- For calculating this, we have considered one of the boundary conditions as:
 - The voltage across the element is zero. ($V = 0$)
- There are 2 more boundary conditions related to force, which are as follows:
 - Unit force normally acts on the right surface in the right direction. ($F_x = 1$)
 - Unit force normally acts on the top surface in the upward direction. ($F_y = 1$)
- After applying the boundary conditions, we will solve for all nodes of the mesh through APDL. On solving, we will get the values of normal and shear stresses and strains at all nodes and then we will calculate the average value of the normal and shear stresses and strains.
- We wrote the average stress and strain values in the form of matrices and through the below formula we calculated the composite stiffness coefficient matrix (C^{comp}).

$$C^{comp} = [\sigma^{av}] [\epsilon^{av}]^{-1} \quad - [1]$$

$$\text{where, } [\sigma^{av}] = \begin{bmatrix} \sigma_x^{av} & \sigma_{xy}^{av} \\ \sigma_{xy}^{av} & \sigma_y^{av} \end{bmatrix}$$

$$[\epsilon^{av}] = \begin{bmatrix} \epsilon_x^{av} & \epsilon_{xy}^{av} \\ \epsilon_{xy}^{av} & \epsilon_y^{av} \end{bmatrix}$$

Calculations:

We need to compare the theoretical value of the coefficient with the experimental value of the coefficients obtained. We started with calculating the value C^{comp} from the experimental data obtained, i.e,

$$\mathbf{1. 40\% Area Fraction} \left(\frac{\text{Area of piezoelectric fibre quarter circle}}{\text{Total area of fibre and matrix}} \right):$$

Total area = Area of square = $10 \times 10 = 100 \text{ sq. units}$

Radius of quarter circle = 7.14 units

Material 1 (PLANE 222) : $\sigma_x = 1.2180, \sigma_y = 1.2182, \sigma_{xy} = 0.0164$

Material 2 (PLANE 182) : $\sigma_x = 0.9068, \sigma_y = 0.9068, \sigma_{xy} = 0.1314$

Average value of materials 1 and 2 :

$$\sigma_x^{av} = 1.0624, \sigma_y^{av} = 1.0625, \sigma_{xy}^{av} = 0.0739$$

$$\begin{bmatrix} \sigma_x^{av} & \sigma_{xy}^{av} \\ \sigma_{xy}^{av} & \sigma_y^{av} \end{bmatrix} = \begin{bmatrix} 1.0624 & 0.0739 \\ 0.0739 & 1.0625 \end{bmatrix}$$

Material 1 (PLANE 222) :

$$\epsilon_x = 0.833 \times 10^{-11}, \epsilon_y = 0.8274 \times 10^{-11}, \epsilon_{xy} = 0.037 \times 10^{-11}$$

Material 2 (PLANE 182) :

$$\epsilon_x = 9.046 \times 10^{-11}, \epsilon_y = 9.045 \times 10^{-11}, \epsilon_{xy} = 5.3 \times 10^{-11}$$

Average value of materials 1 and 2 :

$$\epsilon_x^{av} = 4.9395 \times 10^{-11}, \epsilon_y^{av} = 4.9362 \times 10^{-11}, \epsilon_{xy}^{av} = 2.6685 \times 10^{-11}$$

$$\begin{bmatrix} \epsilon_x^{av} & \epsilon_{xy}^{av} \\ \epsilon_{xy}^{av} & \epsilon_y^{av} \end{bmatrix} = \begin{bmatrix} 4.9395 \cdot 10^{-11} & 2.6685 \cdot 10^{-11} \\ 2.6685 \cdot 10^{-11} & 4.9362 \cdot 10^{-11} \end{bmatrix}$$

Substituting these values in equation [1], we get C^{comp}

$$[C] = \begin{bmatrix} 2.9239 \cdot 10^{10} & -1.4309 \cdot 10^{10} \\ -1.4309 \cdot 10^{10} & 2.9626 \cdot 10^{10} \end{bmatrix}$$

Till now we got the experimental value of $[C^{comp}]$, now, we need to calculate for theoretical values of effective stiffness constant.

$$C_{11}^c = v_f C_{11}^p + v_m C_{11}^m + \frac{v_f v_m}{L} \{ c_1 (C_{12}^p - C_{12}^m) - c_5 (C_{13}^p - C_{13}^m) \}$$

where, v_f = volume fraction of fibre = $\frac{\text{Fibre Area}}{\text{Total Area}} = 0.4$

v_m = volume fraction of matrix = $1 - v_f = 0.6$

$C_{11}^p, C_{12}^p, C_{13}^p, C_{23}^p, C_{22}^p, C_{33}^p$ are the stiffness coefficients for fiber (piezoelectric)

$$C_{11}^p = C_{22}^p = C_{33}^p = 111GPa$$

$$C_{12}^p = C_{23}^p = C_{13}^p = 68GPa$$

$C_{11}^m, C_{12}^m, C_{13}^m, C_{23}^m, C_{22}^m, C_{33}^m$ are the stiffness coefficients for matrix (epoxy)

$$C_{11}^m = C_{22}^m = C_{33}^m = 6.59GPa$$

$$C_{12}^m = C_{23}^m = C_{13}^m = 5.2GPa$$

$$L = (v_m C_{22}^p + v_f C_{22}^m)(v_m C_{33}^p + v_f C_{33}^m) - (v_m C_{23}^p + v_f C_{23}^m)^2$$

$$c_1 = v_m [(C_{13}^p - C_{13}^m)(v_m C_{23}^p + v_f C_{23}^m) - (C_{12}^p - C_{12}^m)(v_m C_{33}^p + v_f C_{33}^m)]$$

$$c_2 = C_{23}^m (v_m C_{23}^p + v_f C_{23}^m) - C_{22}^m (v_m C_{33}^p + v_f C_{33}^m)$$

By substituting the values in all these equations, we get,

$$C_{11}^c = 38.7217 \text{ GPa.}$$

Now, as we have calculated the theoretical as well as experimental value of the effective stiffness coefficient, we can calculate the error as Follows:-

$$Error\% = \left| \frac{X_{Theoretical} - X_{Experimental}}{X_{Theoretical}} \right| \times 100$$

$$\text{Theoretical Value of } C_{11}^c = 38.7217 \text{ Gpa}$$

$$\text{Experimental Value of } C_{11}^c = 29.9239 \text{ Gpa}$$

$$\text{Error: } 29.40\%$$

2. 60% Area Fraction ($\frac{\text{Area of piezoelectric fibre quarter circle}}{\text{Total area of fibre and matrix}}$):

Total area = Area of square = $10 \times 10 = 100 \text{ sq. units}$

Radius of quarter circle = 8.70 units

Material 1 (PLANE 222): $\sigma_x = 1.208, \sigma_y = 1.2275, \sigma_{xy} = 0.0114$

Material 2 (PLANE 182): $\sigma_x = 0.79, \sigma_y = 0.8203, \sigma_{xy} = 0.1049$

Average value of materials 1 and 2 :

$$\sigma_x^{av} = 0.999, \sigma_y^{av} = 1.0239, \sigma_{xy}^{av} = 0.058$$

$$\begin{bmatrix} \sigma_x^{av} & \sigma_{xy}^{av} \\ \sigma_{xy}^{av} & \sigma_y^{av} \end{bmatrix} = \begin{bmatrix} 0.999 & 0.058 \\ 0.058 & 1.024 \end{bmatrix}$$

Material 1 (PLANE 222) :

$$\epsilon_x = 7.52 \times 10^{-11}, \epsilon_y = 7.75 \times 10^{-11}, \epsilon_{xy} = 1.02 \times 10^{-11}$$

Material 2 (PLANE 182) :

$$\epsilon_x = 9.046 \times 10^{-11}, \epsilon_y = 9.045 \times 10^{-11}, \epsilon_{xy} = 5.3 \times 10^{-11}$$

Average value of materials 1 and 2 :

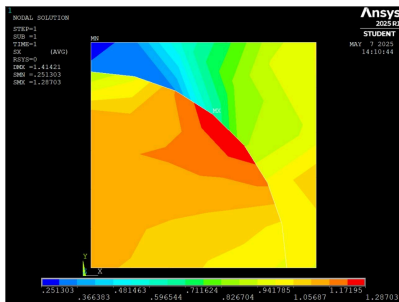
$$\epsilon_x^{av} = 4.456 \times 10^{-10}, \epsilon_y^{av} = 5.697 \times 10^{-9}, \epsilon_{xy}^{av} = 2.486 \times 10^{-10}$$

$$\begin{bmatrix} \epsilon_x^{av} & \epsilon_{xy}^{av} \\ \epsilon_{xy}^{av} & \epsilon_y^{av} \end{bmatrix} = \begin{bmatrix} 4.456 \cdot 10^{-10} & 2.486 \cdot 10^{-10} \\ 2.486 \cdot 10^{-10} & 5.697 \cdot 10^{-10} \end{bmatrix}$$

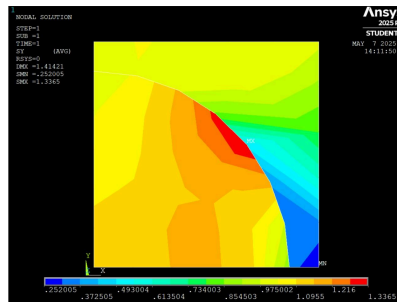
Substituting these values in equation [1], we get C^{comp}

$$[C] = \begin{bmatrix} 2.887 \cdot 10^9 & -1.292 \cdot 10^9 \\ -1.153 \cdot 10^9 & -7.505 \cdot 10^7 \end{bmatrix}$$

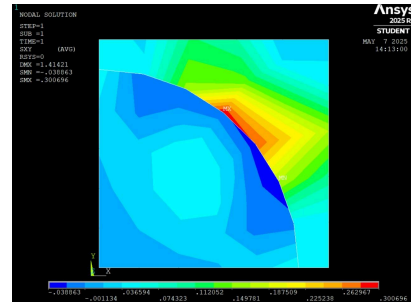
STRESS DISTRIBUTIONS:



X-Component of Stress

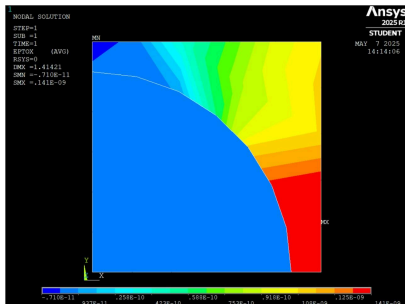


Y-Component of Stress

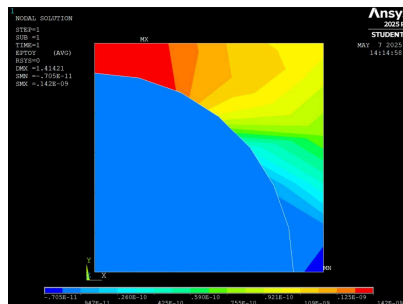


XY-Component of Stress

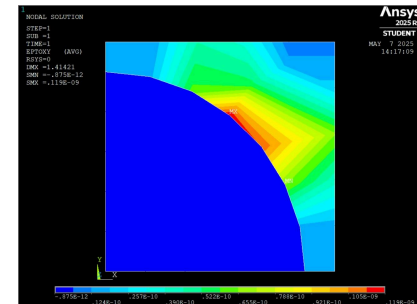
STRAIN DISTRIBUTIONS:



X-Component of Strain



Y-Component of Strain



XY-Component of Strain

3. 80% Area Fraction ($\frac{\text{Area of piezoelectric fibre quarter circle}}{\text{Total area of fibre and matrix}}$):

Total area = Area of square = $10 \times 10 = 100 \text{ sq. units}$

Radius of quarter circle = 9.90 units (approx.)

Material 1 (PLANE 222): $\sigma_x = 1.089$, $\sigma_y = 1.063$, $\sigma_{xy} = 0.046$

Material 2 (PLANE 182): $\sigma_x = 0.902$, $\sigma_y = 0.803$, $\sigma_{xy} = 0.119$

Average value of materials 1 and 2 :

$$\sigma_x^{av} = 0.9955, \sigma_y^{av} = 0.933, \sigma_{xy}^{av} = 0.0825$$

$$\begin{bmatrix} \sigma_x^{av} & \sigma_{xy}^{av} \\ \sigma_{xy}^{av} & \sigma_y^{av} \end{bmatrix} = \begin{bmatrix} 0.9955 & 0.0825 \\ 0.0825 & 0.933 \end{bmatrix}$$

Material 1 (PLANE 222) :

$$\epsilon_x = 6.94 \times 10^{-11}, \epsilon_y = 7.07 \times 10^{-11}, \epsilon_{xy} = 4.88 \times 10^{-12}$$

Material 2 (PLANE 182) :

$$\epsilon_x = 9.45 \times 10^{-10}, \epsilon_y = 8.38 \times 10^{-10}, \epsilon_{xy} = 5.05 \times 10^{-10}$$

Average value of materials 1 and 2 :

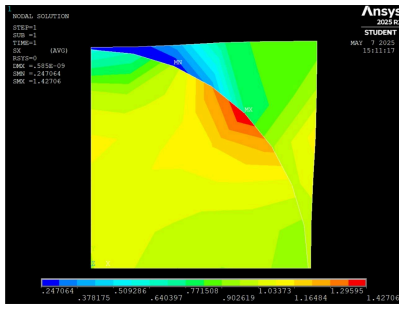
$$\epsilon_x^{av} = 5.072 \times 10^{-10}, \epsilon_y^{av} = 4.543 \times 10^{-10}, \epsilon_{xy}^{av} = 2.549 \times 10^{-10}$$

$$\begin{bmatrix} \epsilon_x^{av} & \epsilon_{xy}^{av} \\ \epsilon_{xy}^{av} & \epsilon_y^{av} \end{bmatrix} = \begin{bmatrix} 5.072 \cdot 10^{-10} & 2.549 \cdot 10^{-10} \\ 2.549 \cdot 10^{-10} & 4.543 \cdot 10^{-10} \end{bmatrix}$$

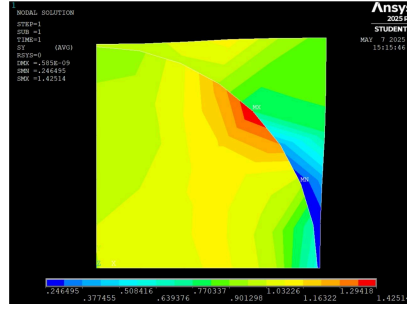
Substituting these values in equation [1], we get C^{comp}

$$[C] = \begin{bmatrix} 2.606 \cdot 10^9 & -1.533 \cdot 10^9 \\ -1.21 \cdot 10^9 & -1.2705 \cdot 10^8 \end{bmatrix}$$

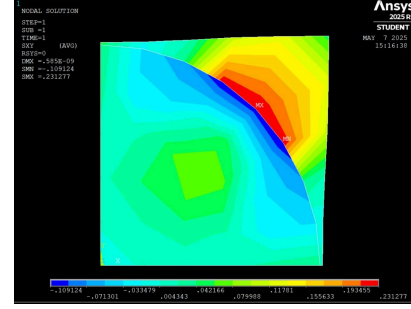
STRESS DISTRIBUTIONS:



X-Component of Stress

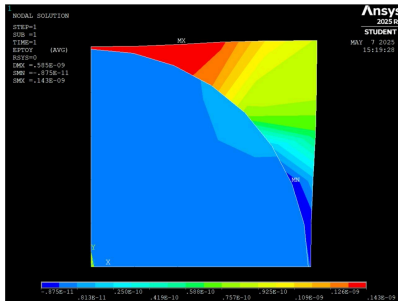


Y-Component of Strain

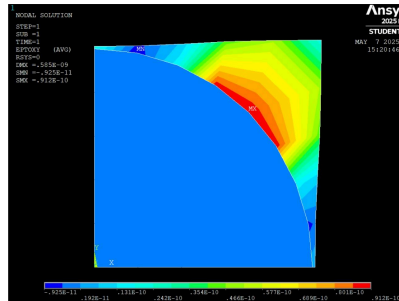


XY-Component of Strain

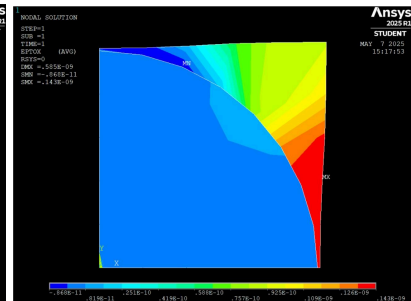
STRAIN DISTRIBUTIONS:



X-Component of Strain



Y-Component of Strain

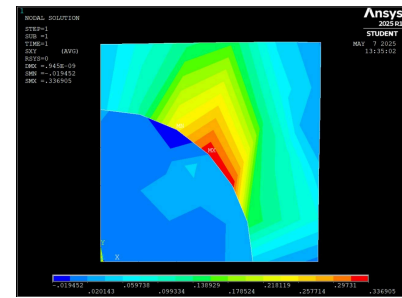
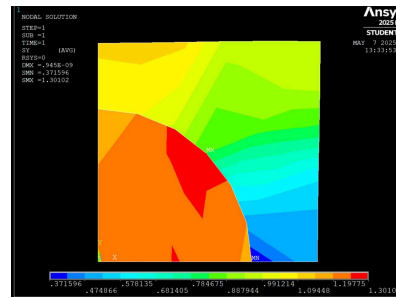
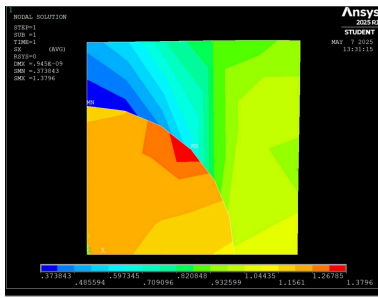


XY-Component of Strain

Observations:

6.11 Stress Analysis:

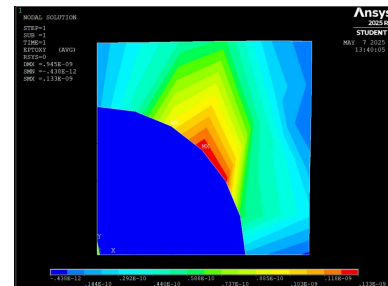
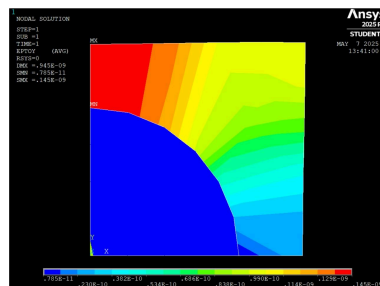
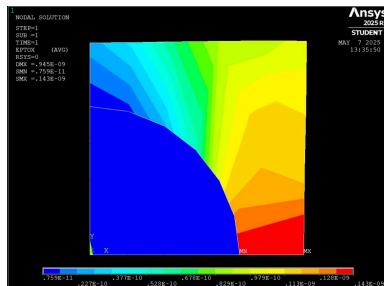
For the sake of simplicity we are going to do the observations only on the cross sectional area element with 40% piezoelectric fibre area. On applying boundary conditions as $F_x = F_y = 1\text{N}$ and $V = 0$, we obtain the following stress distribution.



X-Component of Stress Y-Component of Stress XY-Component of Stress

- When a force F_x is applied to the right edge of the composite square, it generates a non-uniform distribution of normal stress (S_x) throughout the material. As expected, the highest concentration of stress (S_x) is observed near the location where the force is applied, as shown in Figure 1. This region undergoes the most significant load transfer, making it critical in the overall mechanical response of the structure. Both the epoxy and the piezoelectric material in this area are required to work in unison to resist the resulting deformation and maintain structural integrity.
- As we move leftward, the gradual decrease in the x-component of stress (S_x) reflects the diminishing influence of the applied force. This phenomenon can be attributed to the **stress transfer mechanism**, where the concentrated stress at the edge gets progressively distributed throughout the material.
- A similar trend is anticipated for the y-component of stress (S_y). The force application at the top edge is likely to result in the lowest S_y values at the bottommost edge due to the geometry and load transfer pathways.
- The material mismatch between the epoxy and the piezoelectric material plays a significant role in the observed stress distribution. The stiffer piezoelectric material exhibits a higher and more uniform S_x distribution compared to the softer epoxy. This difference in stiffness leads to variations in how each material responds to the applied stress.
- The presence of the cutout and the inserted patch introduces a geometric complexity disrupting the ideal stress distribution.

6.12 Strain Analysis:



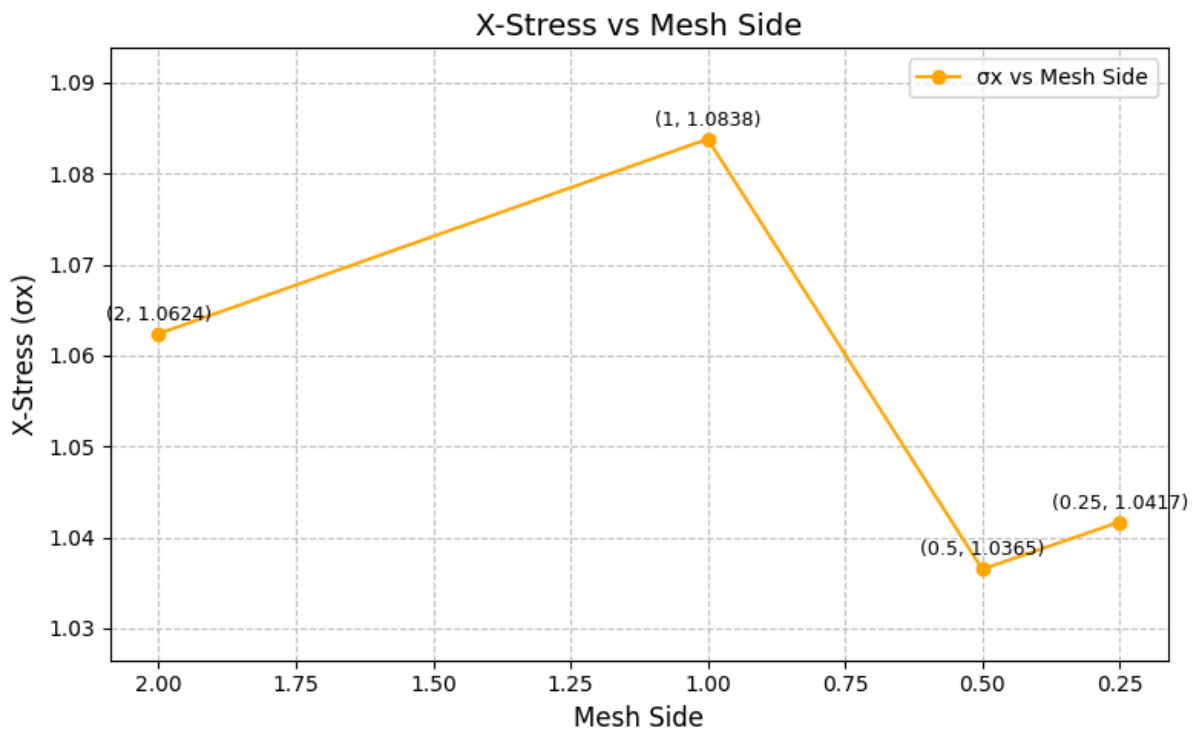
X-Component of Strain Y-Component of Strain XY-Component of Strain

- The first plot, for the X-direction strain distribution, shows that the deformation is least on the left side and most on the right side. This is because the X-direction force is directly acting on the right edge of the geometry and hence its maximum impact will be on the right side, which will reduce on moving towards the left side of the geometry.
- Also, the strain is least in the fiber region because of its relatively higher stiffness than the matrix material.
- Similarly, for the Y-direction strain distribution, shows that the deformation is least at the bottom and most on the top. This is because the X-direction force is directly acting on the top edge of the geometry and hence its maximum impact will be at the top, which will reduce on moving towards the bottom of the geometry.
- For the third plot, where shear strain distribution is shown over the geometry, maximum shear strain is near the middle of the top edge closer to the right side where the combined impact of both forces is maximum, and the minimum shear stress is at the bottom-right and top-left edges where the the impact of one of the forces is negligible and hence least shear deformation occurs. In the region near the interface, the shear deformation is relatively more due to a relatively higher combined impact of forces than in the surrounding regions.

6.2 Graphs:

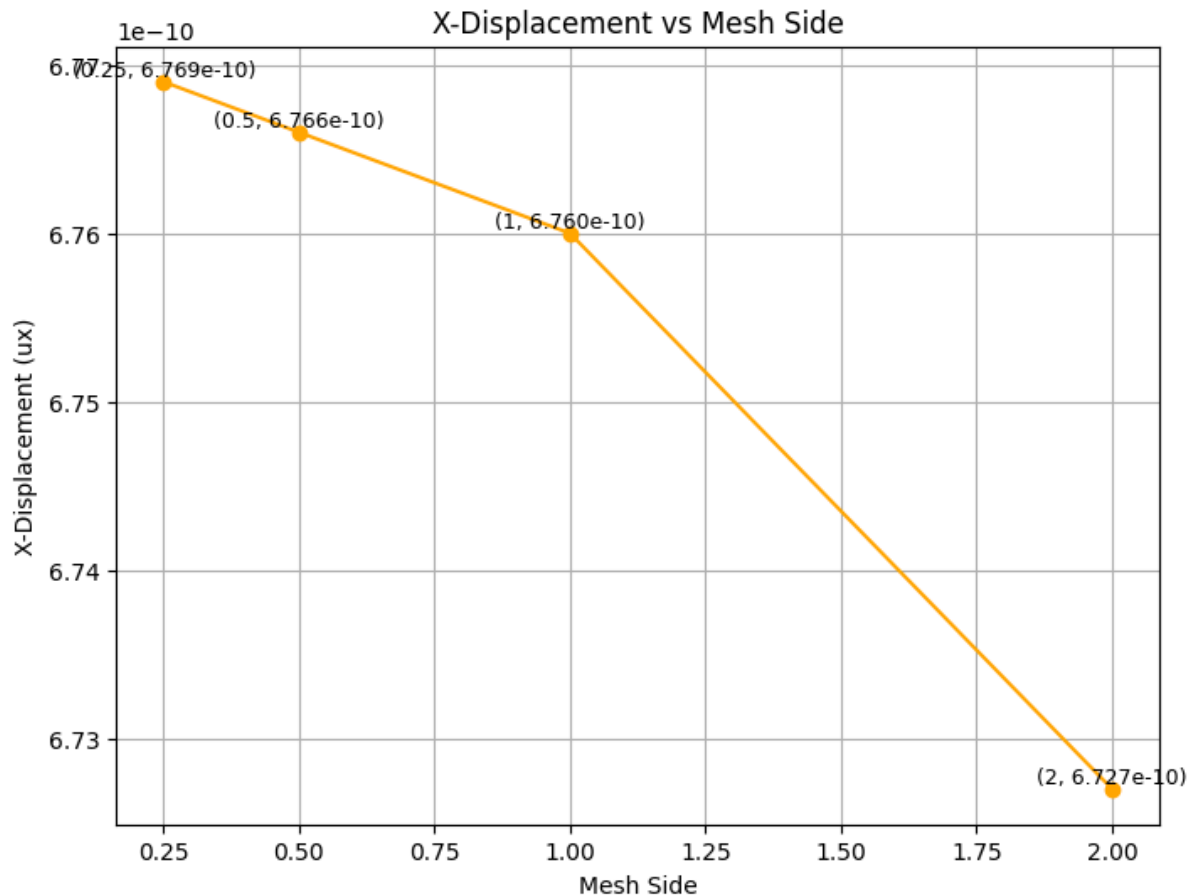
Again for the sake of convenience we have done the analysis only on the area element which has 40% area fraction of piezoelectric material.

6.21 Average stress v/s Mesh side



- We can see from the graph that the average stress in the x-direction of the material does not change much as we refine the mesh by decreasing the size of the mesh side again and again.
- It reaches a stable value of around 1.05 and the maximum variation is only around 4%
- This shows that our solution has **converged** successfully.

6.22 Maximum Displacement v/s Mesh side



- In this graph we have plotted the maximum displacement among all the nodes in the element by varying the mesh side length (by refining the mesh).
- The point chosen for all the mesh sides is the top right corner point as it is impacted the most by both the forces and hence goes through the maximum displacement.
- As with the stress v/s mesh side graph here also we can see that the maximum displacement in the x-direction among all the nodes does not change much when we refine the mesh.

- It reaches a stable value of around 6.7×10^{-10} and the maximum variation is around 0.6% only.
- This again shows that our solution has **converged** successfully.

Results:

The numerical values of the effective properties of the PFRC are evaluated using the method of micromechanics discussed in the preceding section. The elastic and electric properties of the fibers and matrix of the composites are provided to us and are mentioned in the material assignment section.

Following results can be obtained from all the calculations and the plots:

1. The composite stiffness found out by finite element analysis using ANSYS APDL gives us value of C_{11}^c to be 29.9239 Gpa.
2. Whereas the the theoretical value comes out to be 38.7217 Gpa.
3. Thus the variation between the theoretical and experimental values comes out to be around 29%.
4. From the stress distribution graph of 40% area fraction of piezoelectric fibre we can see that the X-direction stress distribution decreases as we go from right to left in the geometry.
5. The stress in the Y-direction also decreases as we go from top to bottom in the geometry.
6. However the shear stress distribution is almost symmetrical about the diagonal axis passing from the bottom left point to the top right point.
7. The stress distribution in the X and Y direction is quite uniform for the piezoelectric fibre and is higher than the matrix.
8. However the shear stress distribution in the piezoelectric matrix is lower than the matrix and also quite uniform.

9. From the strain distribution graph of 40% area fraction of piezoelectric fibre we can see that the X-direction strain distribution decreases as we go from right to left in the geometry.
10. The strain in the Y-direction also decreases as we go from top to bottom in the geometry.
11. The shear strain distribution as well is almost symmetrical about the diagonal axis passing from the bottom left point to the top right point.
12. The strain distribution in the piezoelectric fibre portion is very low and quite uniform as well.
13. The x-stress values in the x-stress vs mesh side graph do not vary much on refining the mesh, showing that the solution converges.
14. The max-x-displacement values in the x-displacement vs mesh side graph do not vary much on refining the mesh as well, showing that the solution converges.

Conclusions:

In this paper, micromechanical analyses have been carried out to determine the effective coefficients of longitudinal PFRC. We considered, the case where $V=0V$, $F_x = 1N$ and $F_y = 1N$, and got to know effective stiffness coefficient, as only mechanical force was applied on the composite. According to some proven formulas (reference Mallik and Ray [2003](#)) theoretical value of effective stiffness coefficient was calculated and compared to the one obtained experimentally. The main objective of the work was to determine the effective stiffness coefficient of the PFRC. Predictions of this coefficient $[C^{comp}]$ by both the cases are in excellent agreement in the useful range of fiber volume fractions.

Related Future Works :

HERE ARE SOME FUTURE WORKS WHICH ARE RELATED TO MICROMECHANICS:

1) Voltage: This project can be extended to applying voltage on the composite to see how the piezoelectric fibre affects the net behaviour of the material.

2) 3D: This project can be extended to three dimensions (3D) also and so the analysis and simulations will be done for its corresponding 3D model.

3) Optimization of Composite Design: Explore optimization techniques to enhance the performance of piezoelectric fiber-reinforced composites. Investigate different fiber orientations, volume fractions, and matrix materials to optimize the mechanical and electromechanical properties of the composite structure. Utilize optimization algorithms to find the optimal combination of parameters for specific applications, such as maximizing energy harvesting efficiency or structural integrity.

4) Novel Composite Materials: Investigate the development of novel composite materials that incorporate advanced piezoelectric elements within innovative matrix structures. Explore unconventional combinations of materials, such as polymers, ceramics, metals, and carbon-based materials, to create composites with enhanced mechanical, electrical, and piezoelectric properties.

5) Multifunctional Composites: Investigate the potential for integrating additional functionalities into the composite material. Explore the design and fabrication of multifunctional composites that combine piezoelectric properties with other functionalities, such as sensing, actuation, or energy storage. This could involve incorporating additional materials or nanostructures into the composite matrix to achieve synergistic effects and expand the range of potential applications.

6) Dynamic Behavior Analysis: Study the dynamic behavior of piezoelectric fiber-reinforced composites under dynamic loading conditions, such as vibrations or impact. Use advanced simulation techniques, such as finite element analysis coupled with modal analysis, to investigate the dynamic

response of the composite structure and optimize its performance for dynamic applications, such as structural health monitoring or vibration damping.

7) Structural Health Monitoring: Explore the use of piezoelectric composites for structural health monitoring applications in aerospace, civil engineering, and infrastructure. Develop innovative sensing and diagnostic techniques based on the electromechanical response of piezoelectric materials to detect and assess structural damage, fatigue, or degradation in real-time.

8) Scale-up and Manufacturing: Explore scalable manufacturing techniques for producing piezoelectric fiber-reinforced composites on a larger scale. Investigate additive manufacturing methods, such as 3D printing, for fabricating complex composite structures with tailored properties. Develop cost-effective manufacturing processes that can be easily scaled up for industrial production while maintaining the desired performance characteristics.

9) Energy Harvesting and Storage: Investigate the potential of piezoelectric composites for energy harvesting and storage applications. Explore the development of piezoelectric nanogenerators that convert mechanical energy from vibrations, motion, or pressure into electrical energy for powering small electronic devices or sensors. Additionally, explore the integration of piezoelectric materials into energy storage devices, such as capacitors or batteries, for enhanced energy conversion and storage capabilities.

10) Bioinspired Materials: Draw inspiration from nature to design bioinspired composite materials with unique properties and functionalities. Explore biomimetic approaches for incorporating hierarchical structures, self-assembly mechanisms, or responsive behaviors into composite materials using piezoelectric elements and matrix materials.

11) Smart Structures and Actuators: Investigate the use of piezoelectric composites as actuators and sensors in smart structures and adaptive systems. Develop advanced control algorithms and feedback mechanisms to optimize the performance of piezoelectric actuators for applications such as active vibration control, shape morphing, or precision positioning in robotics and mechatronics.

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